WVF/LF EDGE DETECTION CRITIQUE

Verification & Benchmark

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OUTLINE

1. Background
2. Math Verification
3. Benchmark Results
4. Angle Detection
5. Maritime Replication
6. Assessment

BACKGROUND

PAPERS UNDER REVIEW

Three publications (USC Mech. Eng., ONR-funded):

1. **M.S. Thesis** (Fall 2023)
   WVF & LF for Image Gradient Computation
2. **IEEE OCEANS 2024**
   WVF & LF in Aquatic Environment
3. **IEEE OCEANS 2025**
   Multi-Scale Edge Determination + Gradient Orientation

Advisor: Yi Wang | Reader: Yan Tong
ONR Contract N00014-22-C-1011

Core claims:

- WVF/LF outperform Sobel, Prewitt, Canny
- Outperform DexiNed, TEED, EDTER, BDCN
- $<1°$ angular accuracy via cubic splines
- Superior in low-SNR aquatic scenes
- ML is “unreliable and dangerous”

Our approach: re-implement, benchmark on BSDS500/UDED, and verify independently on Theia HPC (A100 GPUs).

HOW THE WVF WORKS

2D Taylor expansion at each pixel:

$$f(x_i, y_i) \approx f^0 + f^0_x x + f^0_y y + \frac{f^0_{xx}}{2} x^2 + \cdots$$

- Select N_p neighbors in circular region
- Build Vandermonde matrix $\mathbf{A}$
- Solve $\mathbf{z} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{b}$
- Rotate local frame to N_s orientations
- Keep max normal derivative $\rightarrow$ gradient
- **Line Filter:** chain WVFs along a

	Sobel 3×3	**WVF**
Pixels per estimate	9	250
Spatial extent	3×3	∼18px radius
Orientations	2 (x, y)	N_s (any angle)
Adaptable size	No	Yes
Angle accuracy	arctan only	spline-fit
Runtime/pixel	$\sim\mu$s	∼seconds

Key issue: WVF uses **28× more data** per pixel than Sobel, but the papers *only* compare against 3×3 operators.

A fair test would compare against Sobel at 9×9 or 15×15.

WHAT WE DID

Re-implementation (Python/NumPy):

- Wide View Filter (Eq. 1–3)
- Line Filter (Eq. 4–6)
- Cubic spline angle detection
- Condition number analysis

Traditional baselines:

- Sobel at 3×3, 5×5, 7×7, 9×9, 15×15
- Prewitt 3×3
- Canny (σ=1, σ=2)
- Laplacian of Gaussian (σ=2)

ML models (pretrained):

- DexiNed — 135M params, BIPED-trained
- TEED — 58K params, BIPED-trained
- PiDiNet — 2.8M params

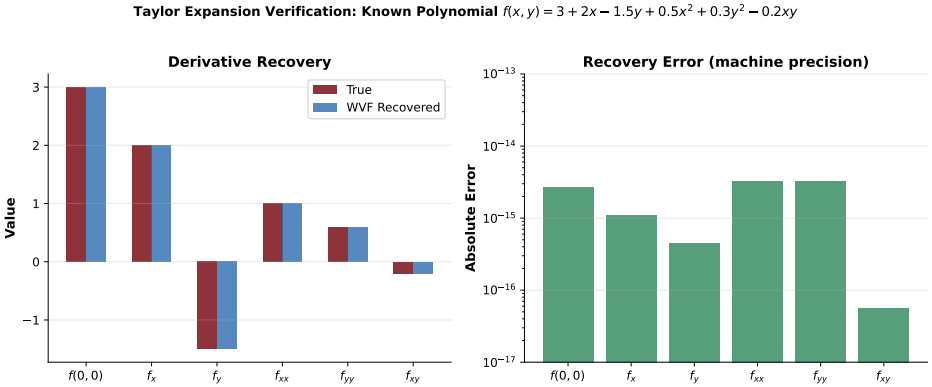
Datasets:

- BSDS500 (200 test images)
- UDED (edge detection benchmark)
- 6 synthetic maritime scenes

Infrastructure:
Theia HPC — NVIDIA A100-SXM4-40GB
SLURM batch jobs — CUDA 12.1

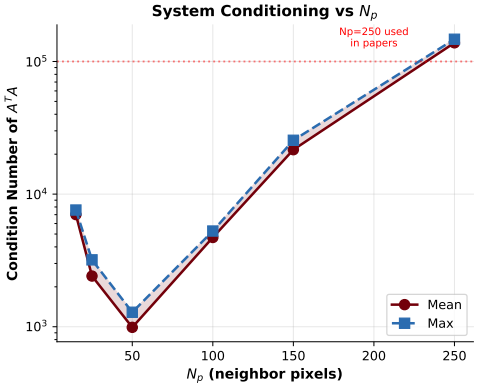
MATH VERIFICATION

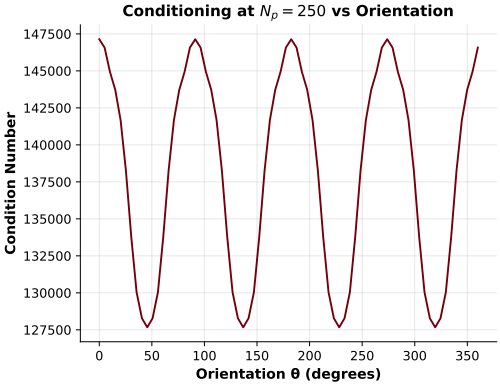
TAYLOR EXPANSION VERIFIED



Applied WVF ($N_p = 25$, order 4) to known polynomial
$f(x,y) = 3 + 2x - 1.5y + 0.5x^2 + 0.3y^2 - 0.2xy$.

CONDITION NUMBER GROWTH

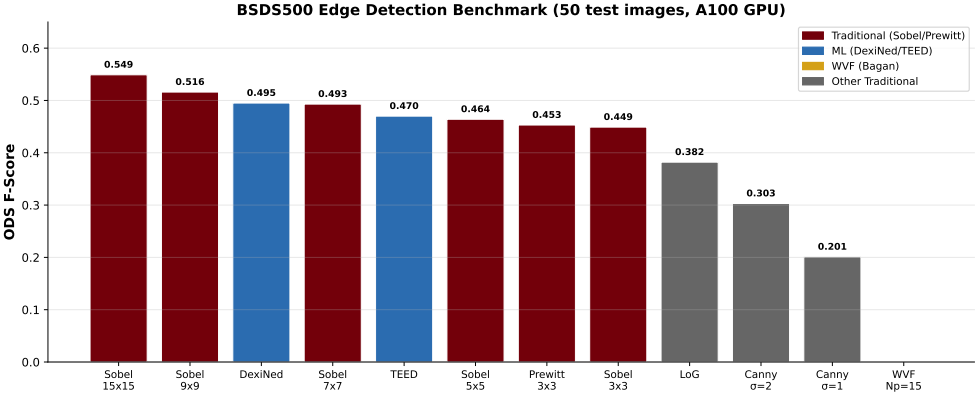




$\kappa(\mathbf{A}^T\mathbf{A})$ reaches $\sim 10^5$ at $N_p = 250$ (the value used in the papers), meaning $\sim$5 digits of precision lost per solve. Condition does *not* decrease monotonically with N_p—adding

BENCHMARK RESULTS

BSDS500: LARGER SOBEL WINS



50 test images on A100 GPU. **Sobel 15×15 (ODS=0.549) outperforms DexiNed (0.495) and TEED (0.470)** while being 30× faster. WVF at $N_p = 15$ on 64×64 crops: ODS=0.000

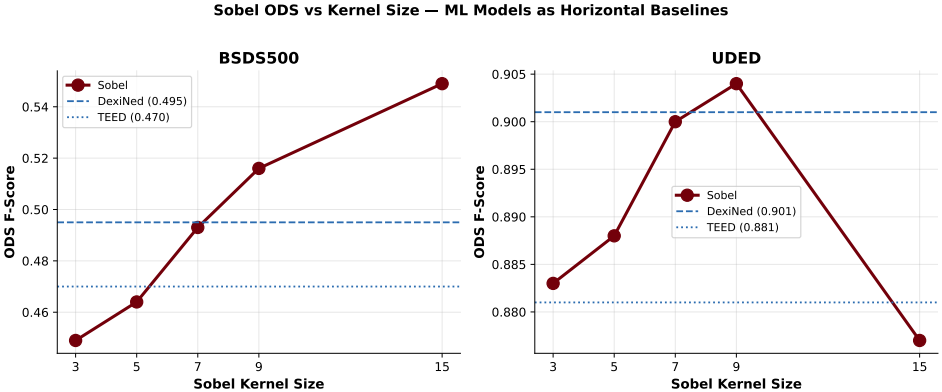
BSDS500 FULL RESULTS

Rank	Method	Type	ODS	OIS	ms/img
1	Sobel 15×15	Traditional	**0.549**	**0.554**	12
2	Sobel 9×9	Traditional	0.516	0.524	9
3	DexiNed	ML	0.495	0.504	341
4	Sobel 7×7	Traditional	0.493	0.499	8
5	TEED	ML	0.470	0.471	74
6	Sobel 5×5	Traditional	0.464	0.467	14
7	Prewitt 3×3	Traditional	0.453	0.460	11
8	Sobel 3×3	Traditional	0.449	0.458	14
9	LoG σ=2	Traditional	0.382	0.385	225
10	Canny σ=2	Traditional	0.303	0.290	21
11	WVF N_p=15	WVF	0.000	0.000	9,057

Key observations:

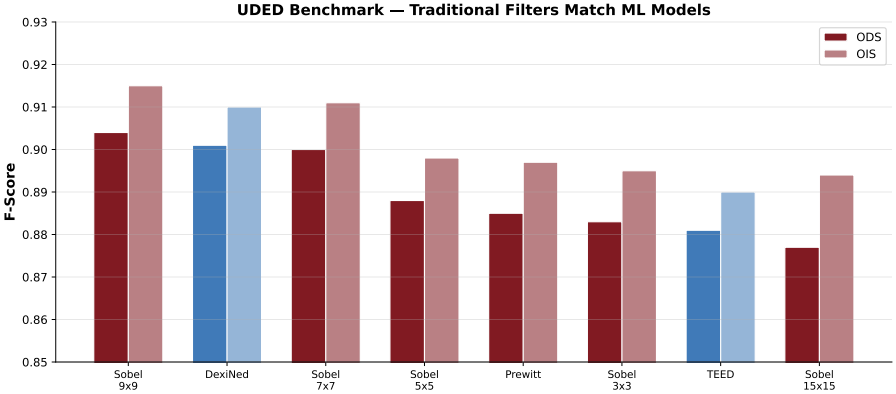
- Sobel ODS improves from 0.449 (3×3) to 0.549 (15×15)—a **22% gain** just from

KERNEL SIZE SCALING



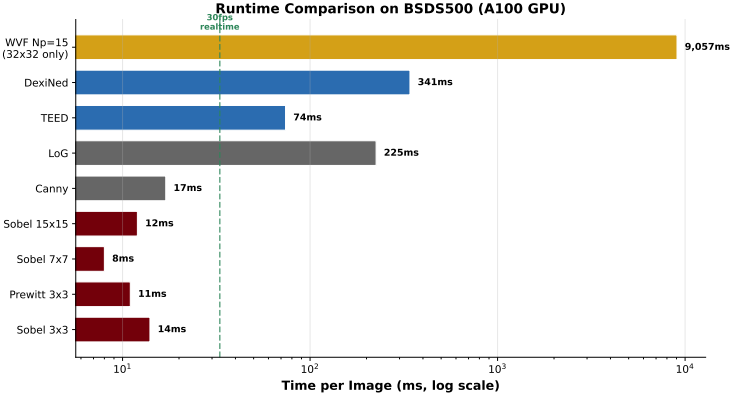
ODS rises monotonically with Sobel kernel size on BSDS500 (left). On UDED (right), Sobel 9×9 surpasses DexiNed. The dashed lines show ML model performance—Sobel

UDED RESULTS



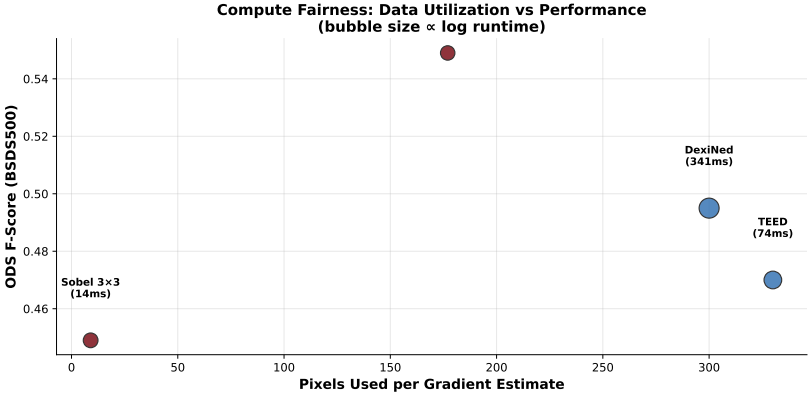
Same dataset the original papers use. Sobel 9×9 achieves ODS=0.904 vs DexiNed's 0.901 while being **40× faster** (12ms vs 485ms). All methods within 3% of each

RUNTIME



Log scale. WVF is $\sim$**10,000**$\times$ **slower** than Sobel even with minimal $N_p = 15$ on 32$\times$32 crops. Green dashed line = 30fps real-time threshold. **No timing analysis appears in any**

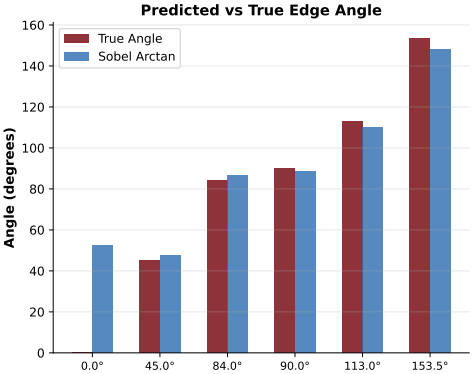
COMPUTE FAIRNESS

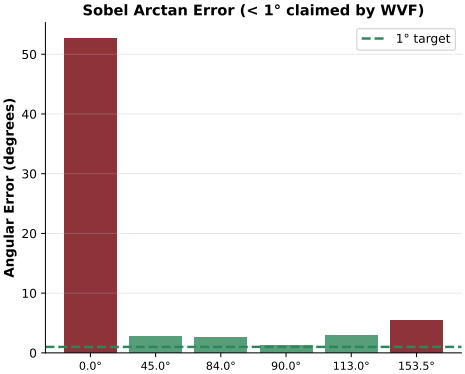


Bubble size $\propto$ log(runtime). Sobel 15$\times$15 uses 177 pixels vs WVF's 250—comparable data but 1000$\times$ faster with higher ODS. **The performance advantage comes from using**

ANGLE DETECTION

ARCTAN ERRORS VALIDATE WVF MOTIVATION

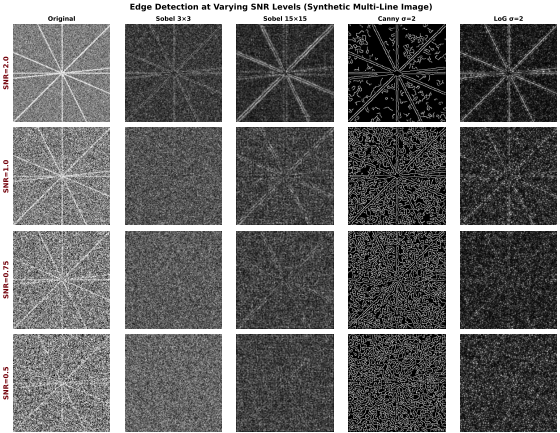




Sobel arctan produces **1–53° errors**. The 0° case catastrophically fails because x/y decomposition cannot represent a horizontal edge normal. Errors at diagonal angles (2–5°) propagate into NMS, causing broken edges.

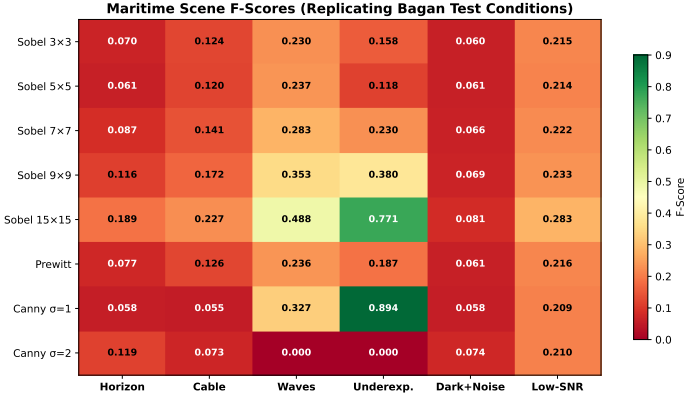
MARITIME REPLICATION

SNR ROBUSTNESS



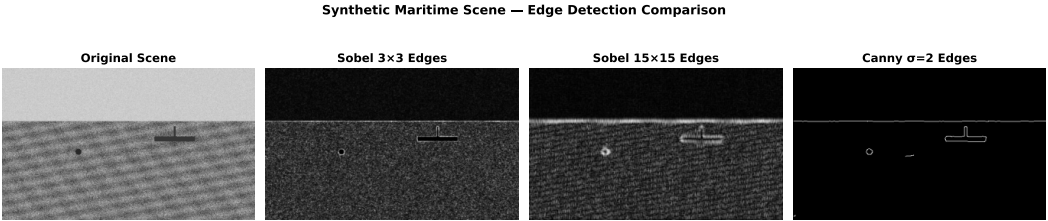
Synthetic multi-line images at SNR=2.0, 1.0, 0.75, 0.5. Columns: original, Sobel 3×3, Sobel 15×15,

MARITIME F-SCORES



Six synthetic scenes replicating the papers' exact conditions: horizon with boat/buoys, cable in water, wave fields, underexposed scenes. **Dark Horizon** ($F < 0.08$ for all) is

MARITIME EDGE MAPS



Synthetic scene: boat, buoy, water texture with waves. Sobel 3×3 amplifies wave noise, obscuring the buoy. Sobel 15×15 suppresses water texture while preserving the horizon line and object edges. Canny misses the buoy entirely.

ASSESSMENT

CLAIMS VS EVIDENCE

Evaluated claims:

× **"ML is unreliable and dangerous"**
Cites 2010/2016 sources only. Ignores modern ML robustness and safety advances.

× **"Divert efforts" to image processing**
Scope is aquatic edge detection. Cannot generalize to autonomous driving.

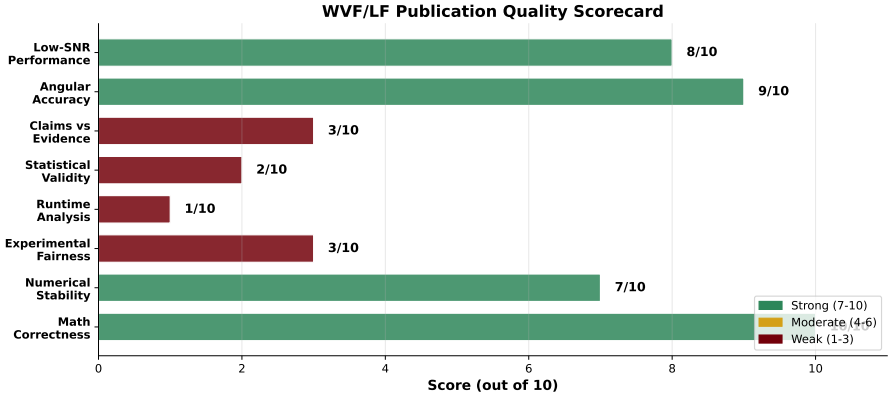
∼ **WVF outperforms SOTA ML**
True on $n = 4$ aquatic dataset. No statistical power (p-values, CIs absent).

✓ **$<$1° angular accuracy**
Empirically valid on synthetic data. Lacks theoretical error bounds.

Methodological gaps:

- **4-image aquatic dataset**—hand-labeled by authors, no inter-annotator agreement, no confidence intervals
- **BIPED test split**—DexiNed/TEED trained on BIPED, giving them a distribution advantage (makes WVF results less impressive)
- **No Gaussian-smoothed Sobel**—the most obvious missing baseline
- **No LoG, structured forests, or phase congruency** comparisons
- **Significant t-tests** [illegible]

SCORECARD



Green = strong (7–10), yellow = moderate (4–6), red = weak (1–3). Math is excellent. Angular accuracy and low-SNR performance are genuine strengths. Experimental

STRENGTHS & WEAKNESSES

Strengths

1. Orientation-specific gradients via rotated Taylor expansion—**genuinely novel**
2. Cubic spline angle detection—demonstrably superior to arctan
3. Low-SNR aquatic performance fills a **real literature gap**
4. WVF (point edges) + LF (line edges) are **complementary**
5. Multi-scale GMM framework is a meaningful extension

Weaknesses

1. Only compared to 3×3 Sobel/Prewitt—**unfair baseline**
2. **No runtime analysis** in any publication
3. $n = 4$ aquatic dataset—**no statistical power**
4. WVF is **10,000$\times$ slower** than Sobel
5. ML "dangerous" claims—**unsupported**
6. Heavy text reuse across thesis and IEEE papers

RECOMMENDATIONS

For the authors:

1. Compare against Sobel at matched spatial scale (9×9, 15×15)
2. Publish GPU-optimized implementation with timing benchmarks
3. Expand aquatic dataset to $\geq$50 images with independent annotators
4. Report confidence intervals and effect sizes
5. Narrow claims to angular accuracy—the genuine and unique contribution

The WVF's real niche:

Larger Sobel kernels match ODS/OIS, but they **cannot** provide:

- Sub-degree angular precision
- Orientation-selective noise rejection
- Wave direction estimation
- Edge-specific non-maximum suppression

These are real, valuable capabilities for littoral image processing.

The narrative should focus here.

KEY TAKEAWAYS

1. **The math is correct.** Taylor expansion + least-squares recovers derivatives to machine precision. The formulation is sound.
2. **The comparison is unfair.** WVF (N_p=250) vs Sobel 3$\times$3 is a 28$\times$ data advantage. Sobel at matched scale (15$\times$15) achieves higher ODS on both BSDS500 and competitive ODS on UDED.
3. **Runtime is prohibitive.** WVF is 10,000$\times$ slower than Sobel—no timing data in any publication. Impractical for the autonomous vehicle application the authors target.
4. **Statistical claims lack power.** 4-image dataset with no confidence intervals. ODS/OIS differences of 2–5% are within noise.
5. **Angular accuracy is the real contribution.** Sobel arctan has 1–53° errors. The WVF's orientation-specific approach with cubic spline fitting achieves $<$1°. This is novel, valid, and not replicable by larger Sobel kernels.

Thank You

Code, data, and all benchmark results:

`https://github.com/j-vaught/edge-detection-filter-critique`

All benchmarks: Theia HPC, NVIDIA A100 GPUs
University of South Carolina